

RaceDay API Endpoint Plan

Authentication					
Method	Route	Description	Role Required	Request Body	Expected Response
POST	/Api/auth/register	Creates a new RaceDay account so a participant or organizer can log in and use the platform.	None (public)	{ fullName, email, password, phone, role }	201 Created - new user and access token. 400 Bad Request - missing/invalid fields. 409 Conflict - email already registered.
POST	/api/auth/login	Verifies a user's email and password and issues an access token so they can reach protected endpoints.	None (public)	{ email, password }	200 OK - access token and refresh token. 401 Unauthorized - incorrect email or password.
POST	/api/auth/refresh-token	Issues a new access token from a valid refresh token, keeping users logged in without re-entering a password.	Any (logged in)	{ refreshToken }	200 OK - new access token. 401 Unauthorized - refresh token invalid or expired.
POST	/api/auth/logout	Invalidates the current session token so the user is securely signed out on this device.	Any (logged in)	None	204 No Content - session ended. 401 Unauthorized - no valid session to end.
POST	/api/auth/forgot-password	Sends a password-reset link to the user's registered email so they can regain access to a locked account.	None (public)	{ email }	200 OK - reset link sent (always returned, even for unknown emails, to avoid revealing which addresses are registered).
POST	/api/auth/reset-password	Sets a new password for the account tied to a valid reset token, completing the forgot-password flow.	None (public)	{ resetToken, newPassword }	200 OK - password updated. 400 Bad Request - reset token invalid or expired.

User profile					
Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the logged-in user's own profile so the app can display their account details.	Any (logged in)	None	200 OK - user profile object. 401 Unauthorized - not logged in.
PUT	/api/users/me	Updates the logged-in user's own contact details.	Any (logged in)	{ fullName?, phone?, email? }	200 OK - updated profile. 400 Bad Request - invalid field values. 409 Conflict - new email already in use.
DELETE	/api/users/me	Deactivates the logged-in user's own account while preserving their historical race records.	Any (logged in)	None	200 OK - account deactivated. 401 Unauthorized - not logged in.
GET	/api/users/{userId}	Lets an organiser or admin look up another user's basic profile, e.g. to verify a participant's identity.	Organiser, Admin	None	200 OK - public profile fields. 403 Forbidden - caller is not an organiser or admin. 404 Not Found - user does not exist.
GET	/api/users/me/history	Returns the logged-in participant's full race history so they can track their personal performance over time.	Participant	None	200 OK - list of past registrations and results (empty array if none yet).

Events					
Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists upcoming and past events, filterable by city, date, or discipline, so participants can browse what's on.	None (public)	None (query params: city, discipline, dateFrom, dateTo)	200 OK - array of matching events (empty array if none match).
GET	/api/events/{eventId}	Returns full detail for one event, including its venue and available routes, so a participant can decide whether to enter.	None (public)	None	200 OK - event detail with nested venue and routes. 404 Not Found - event does not exist.
POST	/api/events	Creates a new event under the organiser's organisation so it can start accepting entries.	Organiser	{ name, eventDate, discipline, description, venueId }	201 Created - new event record. 400 Bad Request - missing/invalid fields. 404 Not Found - venueId does not exist.
PUT	/api/events/{eventId}	Updates an event's details; restricted to the owning organiser so rival organisers can't alter each other's races.	Organiser (owner)	Any updatable event field	200 OK - updated event. 403 Forbidden - not the owning organiser. 404 Not Found - event does not exist.
DELETE	/api/events/{eventId}	Cancels an event, e.g. for weather or safety reasons, so it no longer shows as open for entry.	Organiser (owner), Admin	None	200 OK - event cancelled. 403 Forbidden - not the owning organiser. 404 Not Found - event does not exist. 409 Conflict - event already has finalised results and can't be cancelled.
GET	/api/events/{eventId}/weather	Fetches a live weather forecast for the event's venue and date so participants can prepare for race-day conditions.	None (public)	None	200 OK - forecast temperature, condition, wind, rain chance. 404 Not Found - event does not exist. 502 Bad Gateway - weather provider unavailable.
GET	/api/events/{eventId}/routes	Returns the route(s) and GPX/elevation data for an event so participants can study the course before race day.	None (public)	None	200 OK - array of routes. 404 Not Found - event does not exist.

Categories					
Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists the distance/age/gender categories offered for an event so a participant can pick the one that fits them.	None (public)	None	200 OK - array of categories. 404 Not Found - event does not exist.
GET	/api/categories/{categoryId}	Returns full detail for one category, including its route and entry fee.	None (public)	None	200 OK - category detail. 404 Not Found - category does not exist.
POST	/api/events/{eventId}/categories	Adds a new category to an event, e.g. splitting a marathon into Open and Veteran fields.	Organiser (owner)	{ name, routeId, distanceKm, minAge, maxAge, gender, entryFee }	201 Created - new category. 400 Bad Request - invalid values (e.g. maxAge < minAge). 403 Forbidden - not the owning organiser. 404 Not Found - event or routeId does not exist.

Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/categories/{categoryId}	Updates a category's details, such as correcting the entry fee before entries open.	Organiser (owner)	Any updatable category field	200 OK - updated category. 403 Forbidden - not the owning organiser. 404 Not Found - category does not exist.
DELETE	/api/categories/{categoryId}	Removes a category that's no longer offered, provided nobody has entered it yet.	Organiser (owner), Admin	None	200 OK - category removed. 403 Forbidden - not the owning organiser. 404 Not Found - category does not exist. 409 Conflict - category already has registrations.

Event enrolments (registrations)

Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{categoryId}/registrations	Enters the logged-in participant into a category, creating a pending entry that becomes confirmed once paid.	Participant	{ emergencyContactName, emergencyContactPhone }	201 Created - registration with bib number, payment status "pending". 404 Not Found - category does not exist. 409 Conflict - already entered in this category, or category is full/closed.
GET	/api/registrations/me	Lists all of the logged-in participant's own event entries.	Participant	None	200 OK - array of registrations (empty array if none).
GET	/api/registrations/{registrationId}	Returns detail for one registration; accessible to the participant who owns it or the organiser running that event.	Any (owner), Organiser	None	200 OK - registration detail. 403 Forbidden - not the owner or hosting organiser. 404 Not Found - registration does not exist.
PUT	/api/registrations/{registrationId}	Lets the participant update their own entry details, such as an emergency contact, before race day.	Any (owner)	{ emergencyContactName?, emergencyContactPhone? }	200 OK - updated registration. 403 Forbidden - not the owner. 404 Not Found - registration does not exist.
DELETE	/api/registrations/{registrationId}	Withdraws the participant from an event, freeing their spot in the category.	Any (owner), Admin	None	200 OK - registration cancelled. 403 Forbidden - not the owner. 404 Not Found - registration does not exist. 409 Conflict - event has already taken place.
POST	/api/registrations/{registrationId}/payment	Submits payment for an entry, moving it from pending to confirmed so the participant's spot is secured.	Any (owner)	{ paymentMethod, amount, transactionRef }	200 OK - payment confirmed. 400 Bad Request - amount doesn't match the category's entry fee. 403 Forbidden - not the owner. 404 Not Found - registration does not exist. 409 Conflict - already paid.
GET	/api/events/{eventId}/registrations	Lists everyone entered in an event so the organiser can manage numbers and prepare race-day logistics.	Organiser (owner)	None	200 OK - array of registrations. 403 Forbidden - not the owning organiser. 404 Not Found - event does not exist.

Results

Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Captures a participant's finish result once they cross the line, linking it to their registration.	Organiser (owner)	{ registrationId, finishTime, chipTime, status }	201 Created - result record. 400 Bad Request - invalid time format or status. 403 Forbidden - not the owning organiser. 404 Not Found - registration does not exist. 409 Conflict - a result already exists for this registration.
PUT	/api/results/{resultId}	Corrects a previously captured result, e.g. fixing a timing-chip misread.	Organiser (owner), Admin	Any updatable result field	200 OK - updated result. 403 Forbidden - not the owning organiser. 404 Not Found - result does not exist.
GET	/api/results/{resultId}	Returns detail for a single captured result.	None (public)	None	200 OK - result detail. 404 Not Found - result does not exist.
GET	/api/events/{eventId}/results	Returns the full results list for an event so participants and spectators can see how everyone finished.	None (public)	None	200 OK - array of results ordered by overall position. 404 Not Found - event does not exist.
GET	/api/categories/{categoryId}/results	Returns results filtered to one category so a participant can see how they placed among direct competitors.	None (public)	None	200 OK - array of results ordered by category position. 404 Not Found - category does not exist.
GET	/api/users/{userId}/results	Returns a participant's personal performance history across all events, powering their progress tracking.	Any (owner), Admin	None	200 OK - array of past results across events. 403 Forbidden - profile is private and requester isn't the owner. 404 Not Found - user does not exist.

Planning notes

- Roles assumed: None (public) (unauthenticated), Participant, Organiser, Admin. An Organiser (owner) check means the endpoint also verifies the logged-in organiser's OrganisationID matches the resource's OrganisationID, not just that they hold the organiser role.
- Nesting convention: routes are nested under their parent where the relationship is 1:N and ownership matters for the request (e.g. categories under events), but flattened with an ID once acting on a single specific resource, keeping update/delete URLs shorter.
- Payment is modelled as a sub-action on Registration (POST /registrations/{id}/payment), consistent with the ERD. If a separate PAYMENT entity is introduced later (for refunds or instalments), this becomes its own /api/payments resource.
- Every failure response lists the specific business rule that triggers it (duplicate entries, ownership checks, payment mismatches, deletion conflicts) rather than only documenting the success path.